State-Synchronized Training (SST): Empathic Dialogue as Synchronization in Coupled Affective Dynamical Systems

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Abstract

Empathic dialogue systems built on large language models (LLMs) often produce responses that feel mechanical or formulaic, especially in emotional support and counseling-style interactions. I argue that a primary cause is representational: prevailing approaches treat language as discrete symbols and emotion as static labels or per-utterance scores, while human affect is continuous, multi-dimensional, and dynamically shaped by interaction. Inspired by dimensional emotion theory and dynamical systems, this proposal introduces a new paradigm that models empathic dialogue as a synchronization process between two coupled affective oscillators (user and agent). To operationalize this paradigm, I propose State-Synchronized Training (SST), a computable framework that learns continuous affective state vectors and optimizes not only text likelihood but also (i) real-time affect state inference, (ii) synchronization and supportive guidance of the agent's internal state, and (iii) temporal smoothness of state trajectories. SST aims to endow the model with an intrinsic state-space that can track and regulate affect over turns, producing empathic responses that are more natural, context-sensitive, and dynamically adaptive. The project will validate SST on EmpatheticDialogues and ESConv, with trajectory-level metrics that directly quantify synchronization and regulation quality. By embedding affect dynamics into the model's training objective and interaction mechanism, this work seeks to provide both a theoretical foundation and a prototype system for next-generation affective intelligence.

1. Introduction

LLM-based conversational agents have reached high levels of linguistic fluency, yet they remain noticeably limited in empathic contexts such as emotional support, peer counseling, and mental-health-adjacent conversations. Users commonly describe these systems as "template-like" or "overly generic," even when the content is factually appropriate. This gap is not merely a data problem; it reflects a mismatch between how models represent affect and how affect unfolds in human interaction. In most current pipelines, empathy is treated as an attribute to be recognized or imitated, through discrete emotion categories or one-shot dimensional regression, while the conversation itself is optimized primarily for token-level likelihood. Affect, however, is inherently continuous and time-evolving. Dimensional accounts emphasize that emotional experience varies along continuous axes such as valence and arousal and changes smoothly under contextual and interpersonal influence. In empathic dialogue, effective support is closely tied to co-regulation: the agent must infer the user's affective trajectory, align with it in a validating manner, and gently guide it toward healthier regions without abrupt shifts or "toxic positivity." This behavior is fundamentally dynamical.

This proposal reframes empathic dialogue through a dynamical systems lens: the user and the agent are treated as two interacting affective systems whose internal states evolve over turns. Deep empathy corresponds to adaptive coupling, where the agent both synchronizes with the user's current state and modulates its own trajectory to provide stabilizing, supportive influence. This resembles synchronization phenomena in coupled oscillators, where coordinated behavior emerges from interaction. The central hypothesis is that embedding this perspective into model architecture and training, rather than simulating it only through prompts or external feedback loops, will yield more robust and naturally adaptive empathic responses. To achieve this, I propose State-Synchronized Training (SST), a framework that learns continuous affective state vectors and trains the model to maintain coherent, synchronized state trajectories while generating responses. SST is designed to address an open challenge emphasized by recent surveys: moving toward unified affect understanding and generation with long-horizon affect dynamics in the era of LLMs [1].

2. Related Work

Current approaches to improving empathic responses typically rely on instruction tuning, prompting scaffolds, and feedback-based optimization. Instruction tuning on empathic datasets can improve the frequency of empathic phrasing, but it often amounts to learning stylistic templates and does not explicitly enforce consistent internal affect trajectories across turns. Prompt-based methods can simulate multi-step affect reasoning at inference time, but the "dynamic process" is usually externalized into prompting and does not reliably become an intrinsic, reusable internal state representation. Reinforcement learning and preference optimization introduce goal-directed adjustment, yet the dynamical aspect often resides in the optimization loop rather than in the model's forward-time state evolution. Across these paradigms, empathy is still predominantly treated as a static property or local behavior, not as a coupled dynamical process.

Recent work suggests that representing dialogue with latent state variables and treating emotion-relevant interaction as sequential decision-making can be effective. The SAGE framework demonstrates how state-action augmentation can steer and refine dialogue generation, supporting the broader idea that augmenting generation with structured state representations can improve controllability and coherence [2]. In a parallel direction, RL-EMO formulates multimodal emotion recognition as a sequential decision-making problem, reinforcing the view that affective phenomena are better modeled as trajectories rather than isolated labels [3]. These strands motivate SST, but they do not yet provide a unified training paradigm in which empathic dialogue is explicitly defined as state synchronization between interacting affect systems.

A key gap identified in recent NLP-oriented affective computing surveys is the lack of unified models that jointly understand and generate affect while capturing long-range affect dynamics [1]. Existing approaches may incorporate dynamics indirectly (e.g., via prompting steps or external rewards), but they rarely make continuous state evolution, synchronization, and regulation first-class elements of the model's internal mechanisms and training objective. SST targets this gap by embedding affect dynamics into the learning problem itself.

3. Research Question and Contributions

The central research question is: Can modeling user-agent interaction as a pair of coupled affective dynamical systems, and explicitly optimizing state synchronization and supportive regulation, significantly improve the depth and adaptivity of empathic dialogue responses?

This proposal contributes three elements. First, it formalizes empathic dialogue around a continuous affective state vector that integrates classical emotion dimensions with interaction-relevant intensity or motivational strength, enabling richer representations than static emotion labels. Second, it introduces SST, a training framework that jointly optimizes response generation quality and dynamical objectives: state inference accuracy, synchronization to user state, supportive guidance toward target regions, and smooth state evolution. Third, it proposes trajectory-level evaluation metrics that quantify the *dynamics* of empathy, such as synchronization convergence rate and regulation offset, complementing response-level metrics.

4. Proposed Method: State-Synchronized Training (SST)

4.1 Dialogue as coupled affective state trajectories

Let a conversation unfold over turns t = 1,2,…. SST posits latent continuous affective states for the user and the agent, $\mathbf{s}_t^{(u)}$and $\mathbf{s}_t^{(a)}$. A practical parameterization is a four-dimensional state:

$$\mathbf{s}_t = [v_t, a_t, d_t, i_t] \tag{1}$$

where v denotes valence, a arousal, d dominance, and i an intensity/drive factor reflecting contextual strength (e.g., urgency, insistence, motivational salience). The key is not the exact dimensionality, but the commitment to a continuous, temporally coherent representation that can be inferred and updated at each turn. SST treats empathic interaction as a coupled evolution problem: the agent must infer $\mathbf{s}_t^{(u)}$ from dialogue context, maintain its own $\mathbf{s}_t^{(a)}$, and generate a response conditioned on a planned "ideal" next state $\mathbf{s}_{t+1}^{(a*)}$ that reflects supportive regulation.

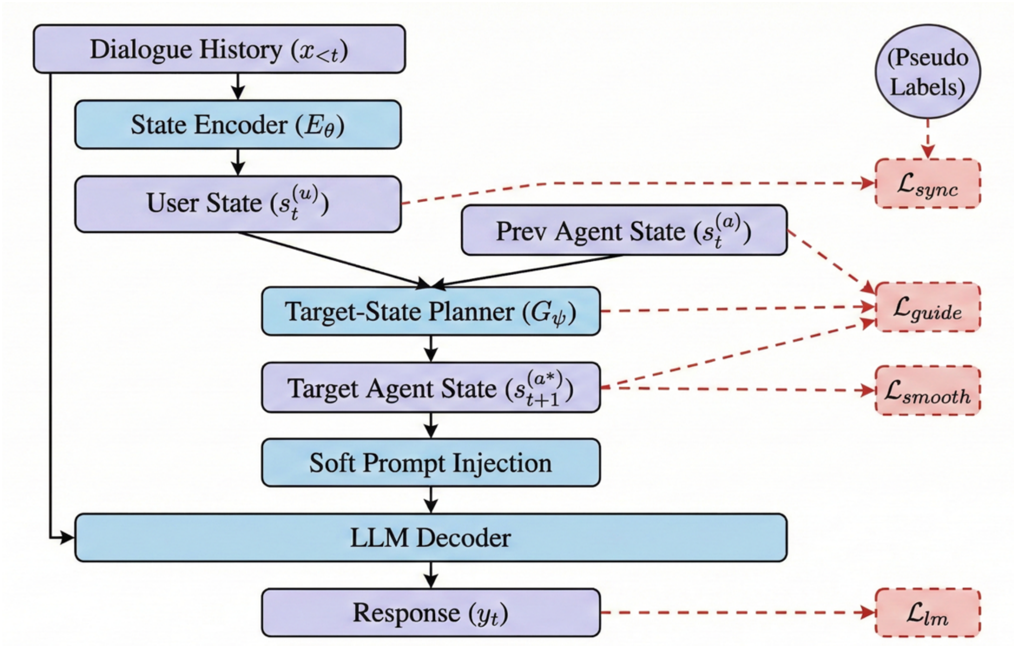

Fig 1. Overview of the SST Framework

4.2 Stage I: Continuous affect state inference with weak supervision

The first stage trains a state encoder E_θ that maps dialogue history to user state estimates:

$$\mathbf{s}_t^{(u)} = E_\theta(x_{\leq t}) \tag{2}$$

Because large-scale datasets rarely provide reliable continuous labels, SST relies on weak supervision and self-supervision. Valence and arousal can be initialized with affect lexicons or pretrained affect models to produce pseudo-continuous signals, while dominance and intensity can be learned from patterns correlated with agency and force in language, such as modality strength, certainty expressions, directive vs. reflective dialogue acts, and interaction context. Training combines a regression objective to pseudo-labels (where available) with auxiliary supervision (e.g., coarse emotion classification) and a temporal smoothness constraint that discourages implausible state jumps. The smoothness prior is essential: without it, the state estimator may overfit to local lexical cues and fail to represent gradual affective change.

4.3 Stage II: Synchronization-aware response generation and target-state planning

In the second stage, SST introduces an interaction mechanism that conditions response generation on affect state. One component is a synchronization-aware attention or conditioning pathway, where state embeddings influence how the model attends to context, encouraging attention patterns consistent with affective alignment rather than purely semantic relevance.

Crucially, SST also introduces a target-state planner G_ψ that predicts an "ideal" next agent state:

$$\mathbf{s}_{t+1}^{(a*)} = G_\psi\left(\mathbf{s}_t^{(u)}, \mathbf{s}_t^{(a)}, x_{\le t}\right) \tag{3}$$

This planner can be implemented as a recurrent state model, a Transformer over state histories, or a continuous-time module (e.g., neural ODE), depending on computational constraints. The planner operationalizes supportive regulation: when the user exhibits high arousal and negative valence, the agent's target state should reflect calm validation and gradual positive shift, rather than abrupt cheerfulness. Finally, the decoder generates the response conditioned on both context and $\mathbf{s}_{t+1}^{(a*)}$. A lightweight state verifier can be used to estimate the implied affective state of generated text, providing a consistency signal that aligns what the agent *intends* (target state) with what it *expresses* (generated response).

4.4 Stage III: SST objective function

SST optimizes a combined objective that preserves language quality while embedding dynamical constraints. The base term is the standard language modeling loss L_{lm}, ensuring fluency and relevance. SST adds three dynamical terms: a synchronization loss L_{sync} that promotes accurate user-state inference and alignment between generated response affect and the target agent state; a guidance loss L_{guide} that shapes $\mathbf{s}_{t+1}^{(a*)}$ to shift in supportive directions relative to $\mathbf{s}_t^{(u)}$ (with bounded, context-appropriate offsets); and a smoothness term L_{smooth} that enforces coherent state trajectories over turns. The final objective is:

$$L = L_{\mathrm{lm}} + \lambda_1 L_{\mathrm{sync}} + \lambda_2 L_{\mathrm{guide}} + \lambda_3 L_{\mathrm{smooth}} \tag{4}$$

This formulation is intended to make "empathy as dynamical co-regulation" a learnable internal structure rather than a surface-level style.

5. Experiments

Experiments will be conducted on EmpatheticDialogues and ESConv, covering both general empathic response settings and emotional support dialogues with longer trajectories. Continuous pseudo-labels for valence, arousal, and dominance will be generated using a combination of lexicon-based estimators and pretrained affect models, with ensembling to reduce bias and noise. The same pseudo-label pipeline will serve as weak supervision for training and as a reference for trajectory-level evaluation, complemented by human evaluation. Baselines will include standard fine-tuning on empathic datasets, prompt-scaffolded inference procedures, and state-augmented dialogue generation approaches inspired by state-action augmentation frameworks. Where feasible, comparisons will also include sequential decision-making formulations for affect modeling that reflect the motivation behind RL-style emotion trajectory learning.

Evaluation will be performed at both the response level and the trajectory level. Response-level metrics will include semantic relevance and standard empathy-related automatic scores, along with distances between generated responses and human references in the learned affect state space. Human evaluation will be conducted as blinded ratings focusing on empathic depth, support effectiveness, dynamic adaptivity across turns, and naturalness (avoiding repetitive templates). Trajectory-level evaluation is central to SST. I will quantify synchronization as a similarity measure between user and agent states over time and examine how quickly the agent re-aligns after affect shifts, producing a synchronization convergence rate. I will also compute regulation offset statistics to test whether the agent's target-state planning is appropriately supportive (bounded, gradual, and context-sensitive), rather than overly aggressive steering.

Ablation studies will remove L_{sync}, L_{guide}, and L_{smooth} in turn, testing whether each component is necessary for improvements in dynamic adaptivity and empathic quality. Long-dialogue analyses on ESConv will visualize state trajectories and synchronization curves to qualitatively validate the "sense–regulate–synchronize" pattern.

6. Timeline

Period (Months)	Key Activities	Primary Deliverables
1-3	- Conduct a focused literature review - Implement the pseudo-label generation pipeline - Reproduce strong baseline models	- Curated related-work map - Pseudo-labeling scripts + documentation - Baseline reproduction results and logs
4-6	- Develop and train the State Encoder - Implement synchronization-aware conditioning (soft prompting) and the target-state planner - Assemble the prototype SST architecture	- Trained state encoder checkpoints - Prototype synchronization-aware module + planner - Initial sanity-check evaluations.
7-9	- Integrate the full SST objective - Train models end-to-end; conduct hyperparameter optimization - Run preliminary automatic evaluation and trajectory analyses	- End-to-end SST model checkpoints - Tuned hyperparameters - Preliminary quantitative results and trajectory visualizations
10-12	- Execute ablation studies and human evaluation - Perform qualitative case studies on ESConv trajectories - Finalize manuscript and prototype release package	- Ablation and human-eval reports - Qualitative ESConv case study set - Submission-ready manuscript - Packaged prototype release

7. Expected Contributions, Risks, and Mitigation

If successful, SST will provide a principled framework that embeds continuous affect dynamics into empathic dialogue modeling, along with a practical training recipe and trajectory-aware evaluation methodology. The primary technical risk is pseudo-label noise for continuous affect supervision. I plan to mitigate this through label ensembling, uncertainty-aware weighting, robust regression losses, and small-scale human calibration. A second risk is over-regularization that could reduce expressive diversity. This will be managed by careful tuning of λ weights and ensuring the language modeling objective remains dominant for fluency and richness. A third risk is inappropriate emotional steering. SST will incorporate bounded guidance constraints and rely on human evaluation for appropriateness, ensuring regulation is supportive rather than coercive.

Overall, this proposal aims to move empathic dialogue beyond static classification and template imitation toward a dynamical, interaction-centered paradigm in which the model learns to track, synchronize, and regulate affect as a continuous process, which is an approach that aligned with the need for unified, long-horizon affect modeling identified by recent surveys.

References

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